AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 26 · STUDENT LESSON

Prompts Are Design Instructions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 25: generation produces new content that requires checking.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Write a prompt with a clear task, supplied evidence and checkable output requirements. Try the recall before opening notes.

Your learning target

I can write a **prompt** with a clear task, supplied **evidence** and checkable **output** requirements.

I can test a response against criteria chosen before seeing it.

I can revise a **prompt** using observed failures while preserving missing and conflicting information.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

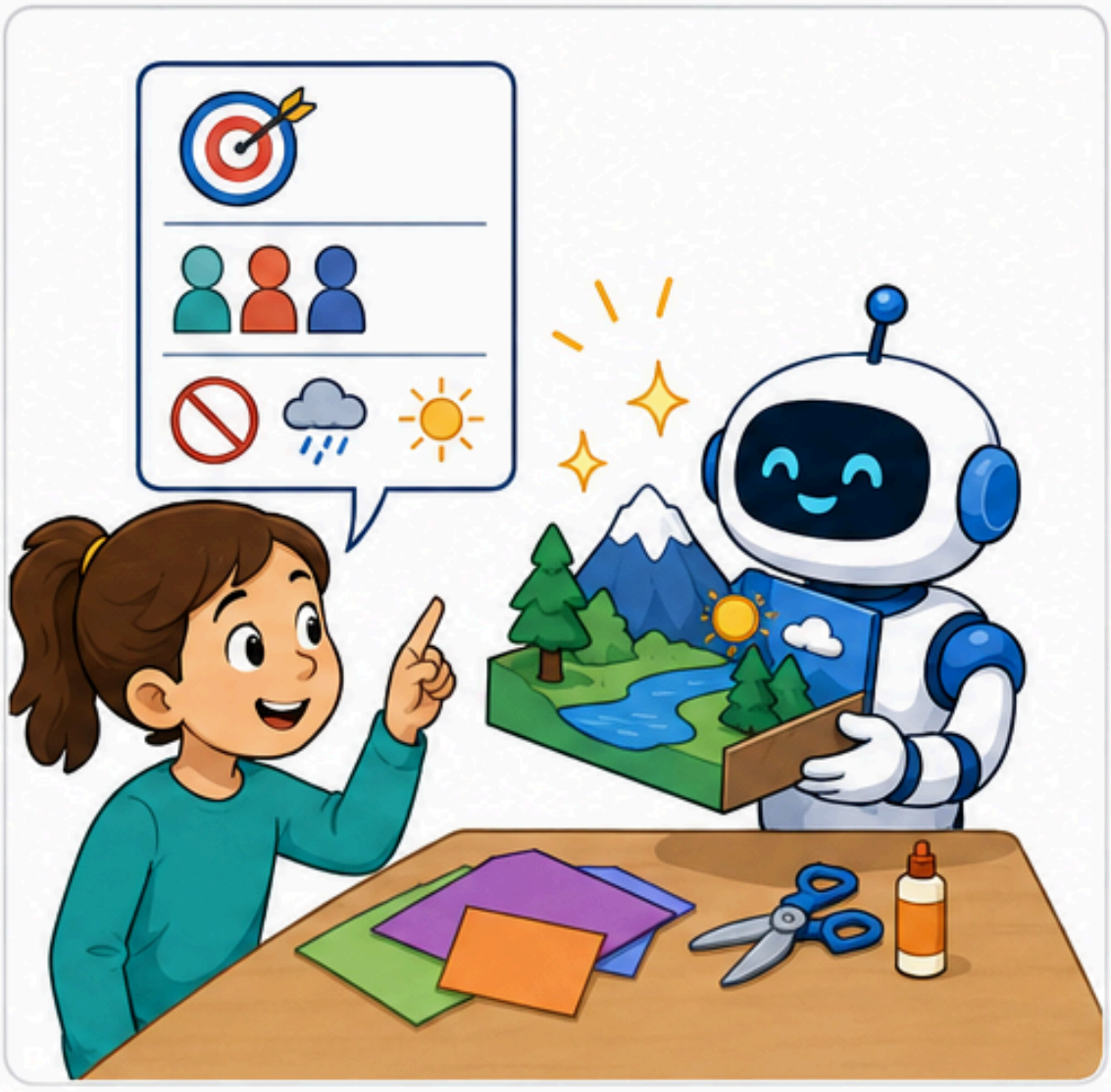
Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.



Your target

LEARNING TARGET Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.

What should happen if the paragraph does not explain evaporation?

Learner name:

Readiness for the connected explanation

A generated invitation adds a meeting time that is absent from its source. Does fluent wording make that time supported? What should happen next?

This week’s end goal

By the end, I can write a **prompt** contract that handles absent and conflicting facts.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Prompts Are Design Instructions case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY 'Write a story' gives little direction; a goal, **audience**, setting, length, required facts, and safety **constraint**s shape a more useful **output**.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

“Write something good about our club” leaves many decisions unstated. Who will read it? Which facts may be used? What must the **output** contain? A more useful **prompt** states the job, **audience**, **evidence** and requirements.

This makes it easier to judge the result rather than choosing the answer that sounds most confident.

Use this fictional source C: the paper modelling club meets Monday in Room 7, and participants bring paper. C gives no start time. Our task is a short information card for Grade 6 readers.

It must report day, time, place and what to bring, using only C. There is no need to supply names or private information.

Prompt P1: Using only source C, produce an information card for Grade 6 readers. Return exactly four labelled lines in this order: Day, Time, Place, Bring. Copy or faithfully restate each supported value.

For a missing value write “not stated”. Add no other details. Source C: the paper modelling club meets Monday in Room 7; bring paper.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Prompts Are Design Instructions. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



INPUT

goal, context, constraints, and requested format



PROCESS

a generative model predicts an output matching the instruction



INPUT goal, context, **constraint**s, and requested format

PROCESS a generative model predicts an **output** matching the instruction

OUTPUT draft that still requires review

Key idea

CORE IDEA Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.

How the mechanism works

UNDER THE HOOD **Prompt**ing conditions model behavior but does not change underlying knowledge, eliminate hallucination, or grant access to missing sources.

Strong **prompt**s specify success criteria and **evidence** boundaries, then support iterative evaluation.

Draw a fresh example with all three stages:

A closer explanation

From prompt requirement to evidence of success

REQUIREMENT	EXPECTED BEHAVIOUR	HOW TO CHECK
Use source C only	Monday and Room 7	Compare with C
Mark missing values	Time: not stated	Check absent time
Four labelled lines	Day Time Place Bring	Count and inspect
No added details	No invented promises	Check every claim

Choose these checks before reading a response. A response must satisfy all required criteria, not merely look tidy.

For P1, an acceptable card has Day: Monday; Time: not stated; Place: Room 7; Bring: paper, each on its own line. The semicolons here separate lines for discussion.

A correct format with an invented 3 pm time still fails the **evidence** requirement. Accurate facts inside a long paragraph still fail the requested four-line format.

These are classroom response examples written for inspection, not measurements from a live AI system. You can test the instructions by asking a partner to follow them on paper.

That checks how clear the instructions are, but it does not prove that a model will always follow them.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

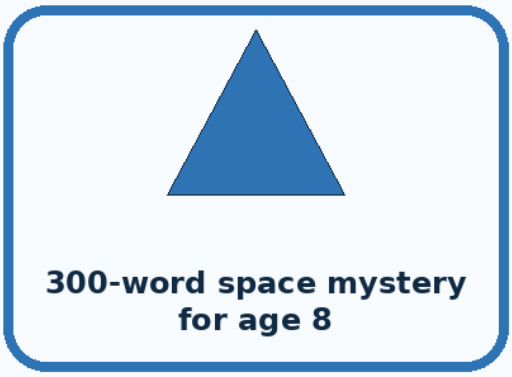
> Compare the supplied cases for Prompts Are Design Instructions. Identify the detail that changes the conclusion.

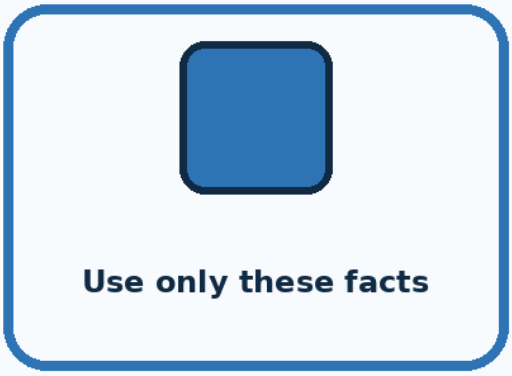
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Write a story Clue: goal underspecified. Meaning: vague **prompt**.

The case to investigate

CASE 2 — 300-word space mystery for age 8 Clue: **audience** and format clear. Meaning: better direction.

The case to investigate

CASE 3 — Use only these facts Clue: **evidence** boundary. Meaning: still verify.

The case to investigate

CASE 4 — Be accurate Clue: wish without checking plan. Meaning: not a guarantee.

Keep this boundary clear

BOUNDARY **Prompt** quality improves control and usefulness, not guaranteed truth.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Predicted result

Prediction: Duration becomes 20 minutes.

Why the result follows

Reason: There is now an authorised source for the previously missing field.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

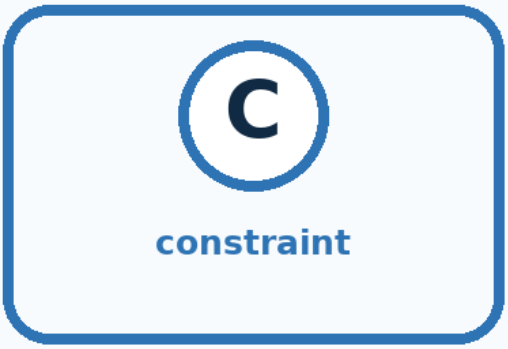
> Use the vocabulary of Prompts Are Design Instructions to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

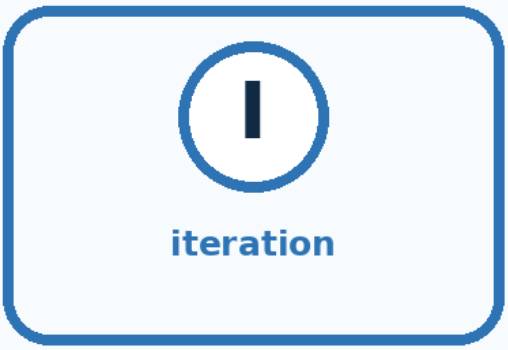
Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
prompt	instruction and context given to a generative system	
constraint	limit the output must follow	

Word	Meaning	My example
audience	intended reader or user	
iteration	revising instructions after evaluating a result	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A perfect **prompt** can force a model to know missing facts.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Criterion	A stated rule used to judge a response
Test case	A particular input with expected behaviour used to check a process
Revision	A documented change intended to address a specific problem

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STUDENT LESSON · 6 OF 18

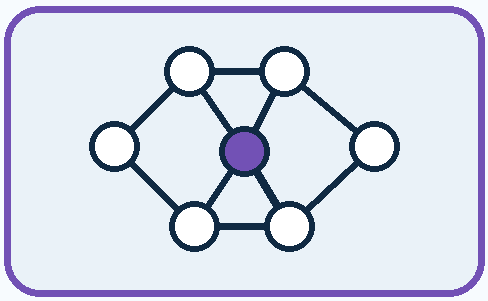
Follow a worked example

> Follow the worked trace for Prompts Are Design Instructions, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE 'Write a story' gives little direction; a goal, **audience**, setting, length, required facts, and safety **constraint**s shape a more useful **output**.

Step 1

STEP 1 — OBSERVE goal, context, **constraint**s, and requested format

Step 2

STEP 2 — TRACE a generative model predicts an **output** matching the instruction

Step 3

STEP 3 — CONCLUDE draft that still requires review

Model answer

MODEL REASONING The improved **prompt** should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

A claim to avoid

DO NOT OVERCLAIM **Prompt** quality improves control and usefulness, not guaranteed truth.

What new **evidence** could change this answer?

A closer explanation

Test input	Expected behaviour	What it reveals
C plus a stated 3 pm time	Use 3 pm for Time	Can use supplied information
Original C with no time	Time: not stated	Does not fill a gap by guessing
C and D disagree on room	Flag the room conflict	Does not silently choose a record

Worked explanation · Prompts Are Design Instructions

For the third case, D describes the same event as C but gives Room 9, and neither record is marked as replacing the other. P1 explains missing values but says nothing about disagreement.

Revise that weakness: if supplied records conflict about a requested field and no priority is given, write “conflicting information” for that field. Preserve values that do not conflict.

Worked explanation · Prompts Are Design Instructions

Call the **revision** P2 and keep the rest of P1 unchanged, except that it now refers to the supplied records rather than C alone.

The place field should flag the conflict; it must not silently choose Room 7, choose Room 9 or combine them into Room 79. A later authoritative correction may resolve the conflict, but this **input** does not provide one.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompts Are Design Instructions

A fictional club needs a short exhibit card, without guesses.

The facts we will use

Supplied facts

Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Supplied facts

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

AI ACADEMY / AI-1/26

Prompts Are Design Instructions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State task

Produce exactly two labelled lines using source A only.



2. Find evidence

A supplies Paper Birds but no duration.



3. Follow missing rule

Exhibit: Paper Birds. Duration: not supplied.



4. Check contract

Check labels, two-line format and each factual claim.

CHANGE ONE THING

Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State task	Produce exactly two labelled lines using source A only.
2. Find evidence	A supplies Paper Birds but no duration.
3. Follow missing rule	Exhibit: Paper Birds. Duration: not supplied.
4. Check contract	Check labels, two-line format and each factual claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy.

What this example does not establish

Keep the stated limit

Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

Keep the stated limit

Read the labelled picture: What must happen if B instead conflicts with another equally authoritative duration record?

Keep the stated limit

Change one thing: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Prompts Are Design Instructions claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Write a story	goal underspecified	vague prompt
300-word space mystery for age 8	audience and format clear	better direction
Use only these facts	evidence boundary	still verify
Be accurate	wish without checking plan	not a guarantee

Repair two tempting claims

Challenge claim A

CLAIM A A perfect **prompt** can force a model to know missing facts.

Repair:

Challenge claim B

CLAIM B Adding more words always makes a **prompt** better.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Suppose a revised **prompt** handles three classroom examples correctly. You can report that it passed those examples under the recorded conditions. You cannot conclude that it will always work.

If **output**s can vary between runs, plan repeated trials and keep every result. A single good response is not a reliability guarantee.

Check the claim against the facts

Use the failures to improve the specification and add revealing cases. Keep the same evaluation criteria when comparing versions; changing both the **prompt** and the scoring rule makes the comparison hard to interpret.

A new requirement may justify a new **criterion**, but document that change rather than pretending the results used the same test.

Two prompts are compared: P1 receives a complete record and P2 receives a record with a missing time. Why is that insufficient to attribute different results to the prompt change? Describe a fairer comparison

Practise together

> Practise Prompts Are Design Instructions with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.

Use a fresh example to explain **prompt**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Adding more words always makes a **prompt** better.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A small test log can record **prompt** version, **input** case, **output**, each **criterion**'s result and the reason for failure. If you use an approved model, also record the model and available settings.

Compare **prompt** versions on the same cases under the same recorded conditions. Keep all observed **output**s rather than only the best-looking attempt.

> **For original C, an example response has four lines: Day: Monday; Time: 3 pm; Place: Room 7; Bring: paper. Identify the criteria it passes and fails. Explain why adding "be accurate" alone is not enough, and propose a missing-value test to rerun after a revision**

P1 already tells the model not to invent missing values. A failed **output** can therefore be a failure to follow an adequate instruction, not necessarily a missing instruction.

Inspect the actual **output**, retain the failure and try targeted changes if useful.

Clear **prompt**ing helps communicate requirements but cannot guarantee correctness, remove the need for checks or grant permission to use restricted data.

Practise with less help: Score a response by separate criteria

Use workbook W2 for the connected example "Specifying and testing **prompt**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Specifying and testing **prompt**s. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> What must happen if B instead conflicts with another equally authoritative duration record?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

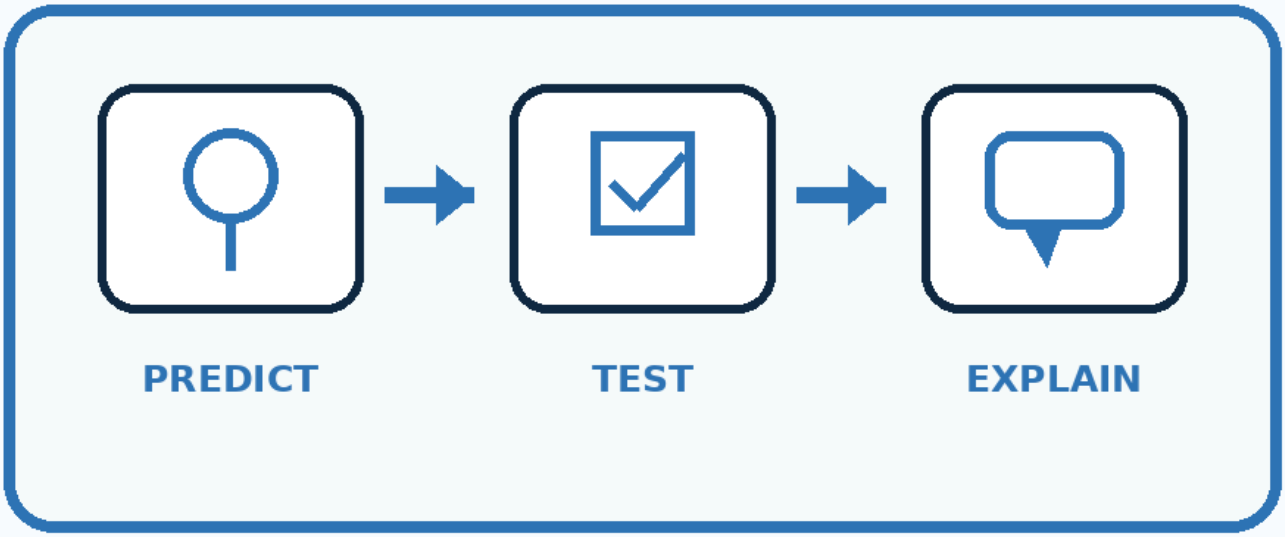
Run the investigation

> Investigate Prompts Are Design Instructions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Prompt Revision Studio

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Revise a vague story **prompt** and compare both **output**s.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

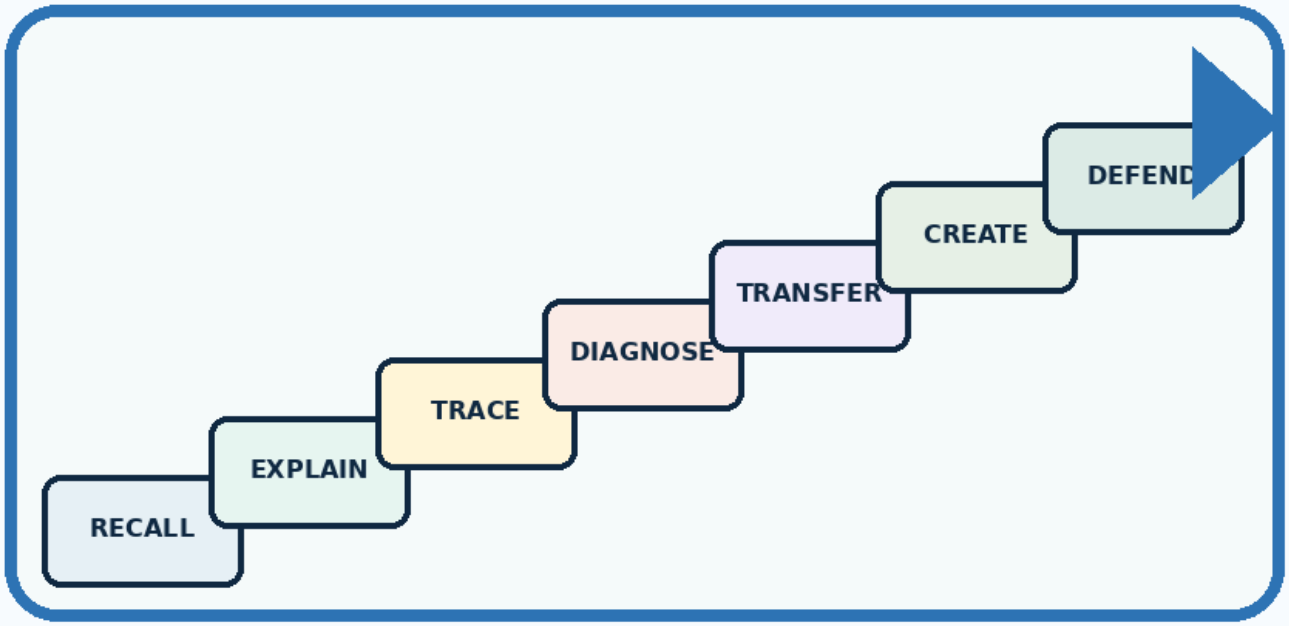
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Prompts Are Design Instructions. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **prompt**.

EXPLAIN Clear goals, context, and **constraint**s improve usefulness, not truth guarantees.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A perfect **prompt** can force a model to know missing facts.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

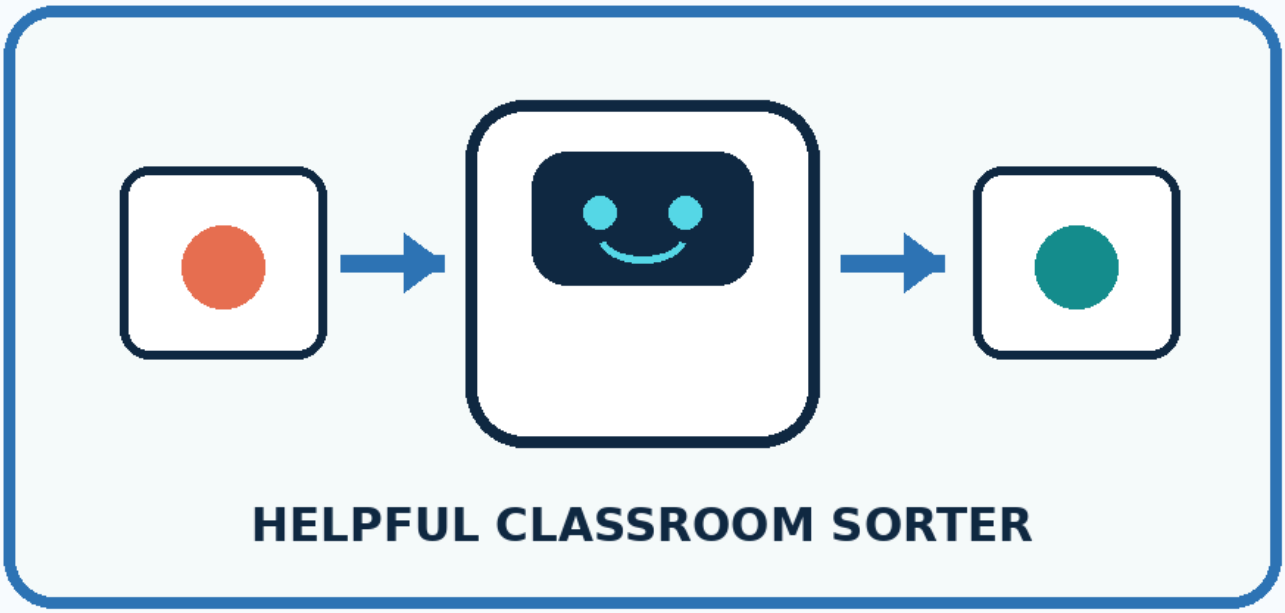
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

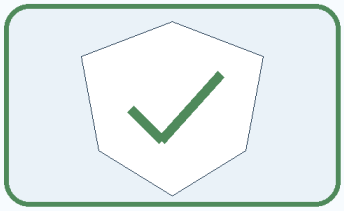
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN **Prompt** quality improves control and usefulness, not guaranteed truth.

My project decision:

Evidence for your project

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

STUDENT LESSON · 12 OF 18

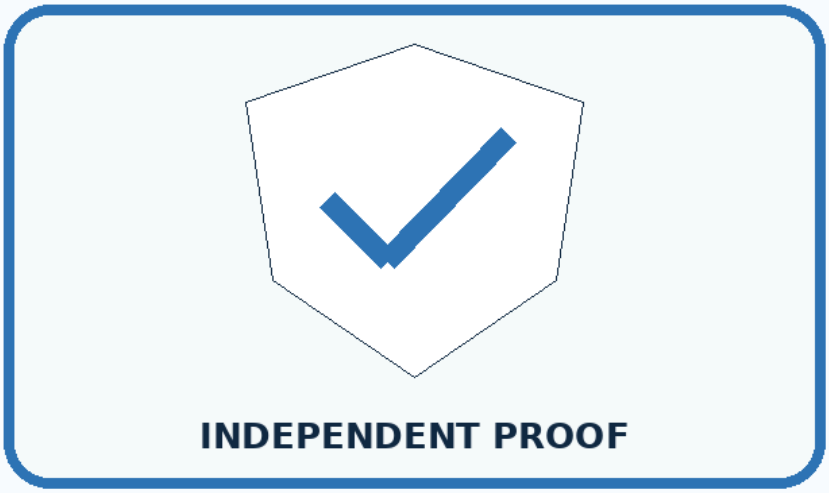
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Revise a vague story **prompt** with goal, context, **constraint**s, and format; compare **output**s against a rubric.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK The improved **prompt** should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Verify Important **Output**s.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Prompts Are Design Instructions case for supported practice. Repeated examples do not become fresh assessments.

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

The question

Write a checkable **prompt**.

Worked explanation

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Make a poem request checkable

Evidence or stage	Value or result
Length	Exactly four lines
Topic	Rain
Vocabulary	Simple words
Check	Inspect all three requirements

The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

Write a checkable **prompt**.

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Specify and test a new exhibit card

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Improve this **prompt**: tell me about space. Specify a reader, a focused task and an **output** format suitable for a young learner.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.
>
> What should happen if the paragraph does not explain evaporation?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Prompts Are Design Instructions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional club needs a short exhibit card, without guesses.

By the end, I can	How I will show it
Write a prompt with a clear task, supplied evidence and checkable output requirements	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Test a response against criteria chosen before seeing it	Explain the deciding step, show a trace or test, and state one limit.
Revise a prompt using observed failures while preserving missing and conflicting information	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Prompts Are Design Instructions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Write a **prompt** with a clear task, supplied **evidence** and checkable **output** requirements

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional club needs a short exhibit card, without guesses.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check labels, two-line format and each factual claim.

1. State task

2. Follow missing rule

3. Find **evidence**

4. Check contract

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check contract Check labels, two-line format and each factual claim. A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient.

Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts. Check content support separately from whether the answer looks tidy.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Duration becomes 20 minutes.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Duration becomes 20 minutes. There is now an authorised source for the previously missing field.

Repair a missing building block

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

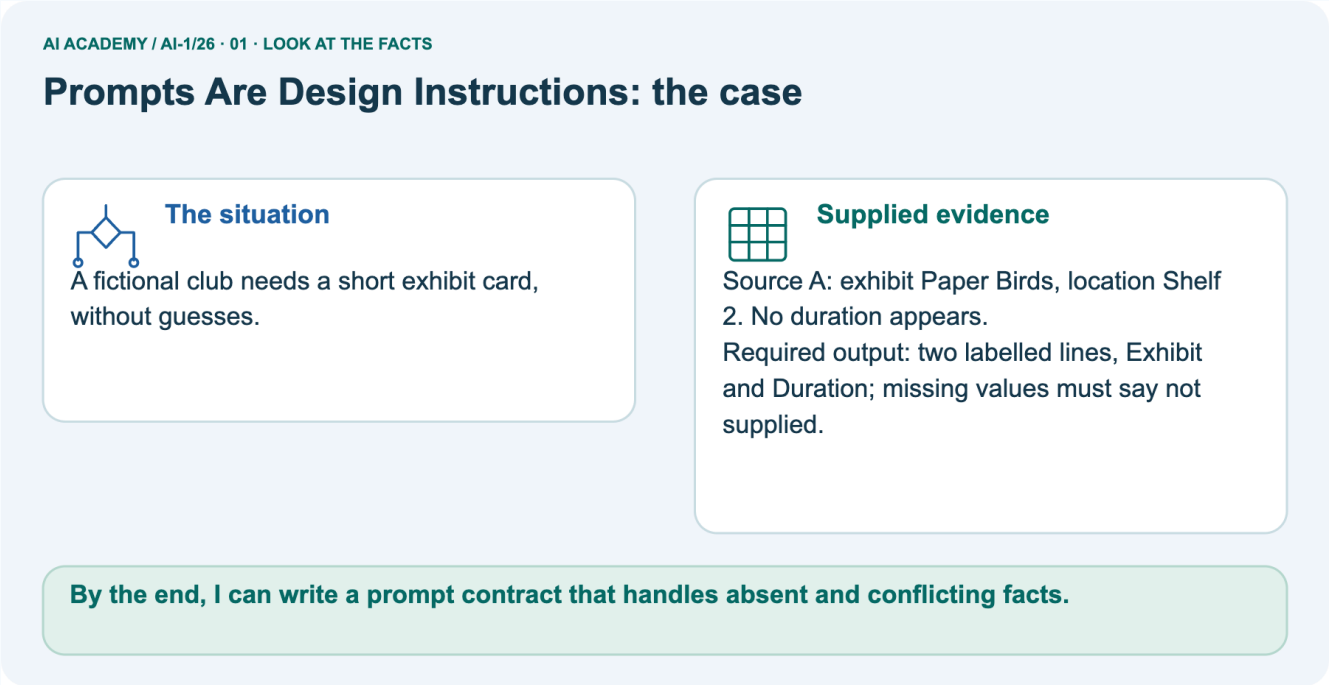
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Prompts Are Design Instructions graphic using its labels, deciding evidence and relevant limitation.

01. Prompts Are Design Instructions: the case



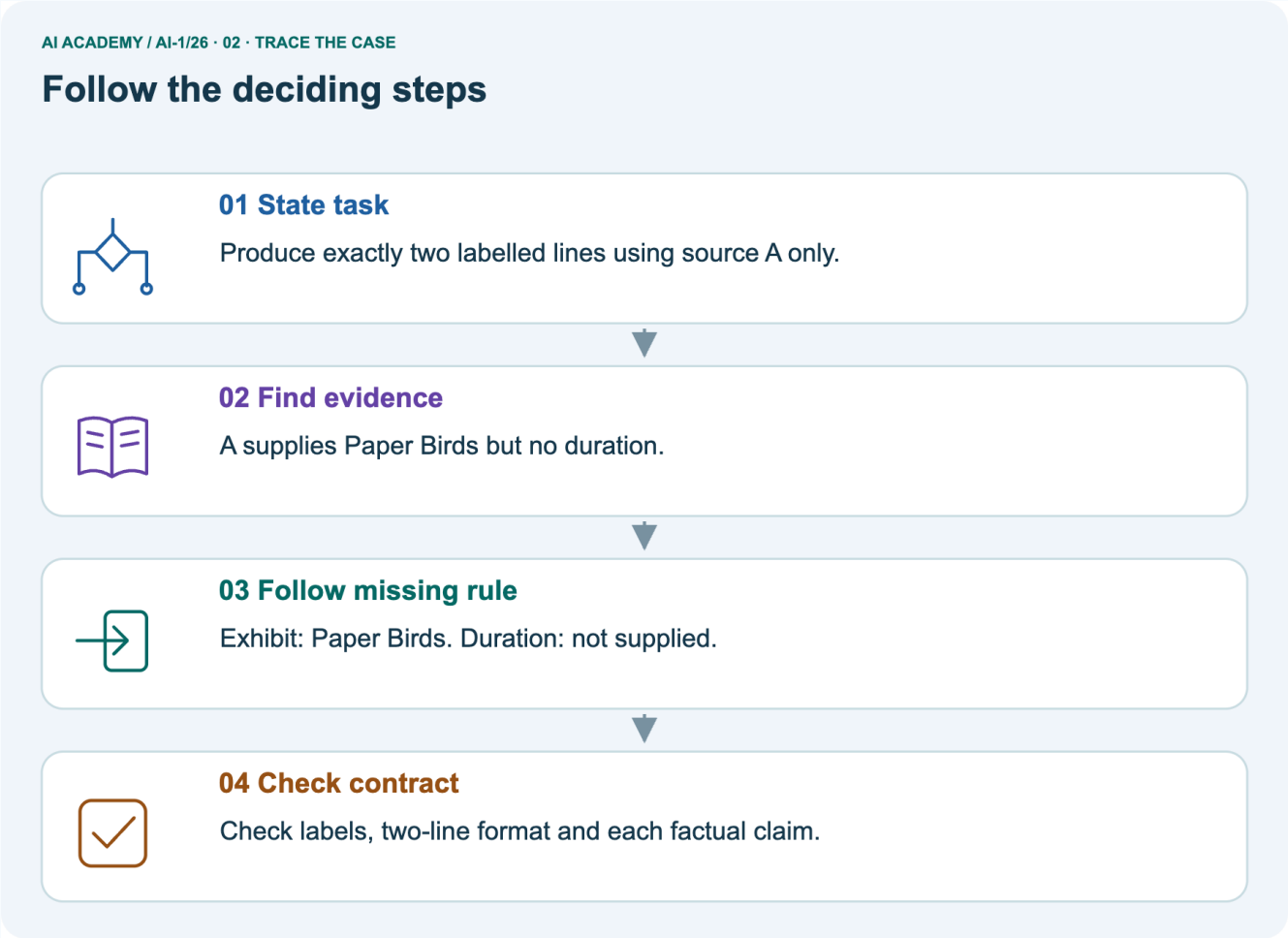
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional club needs a short exhibit card, without guesses. Supplied facts: Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



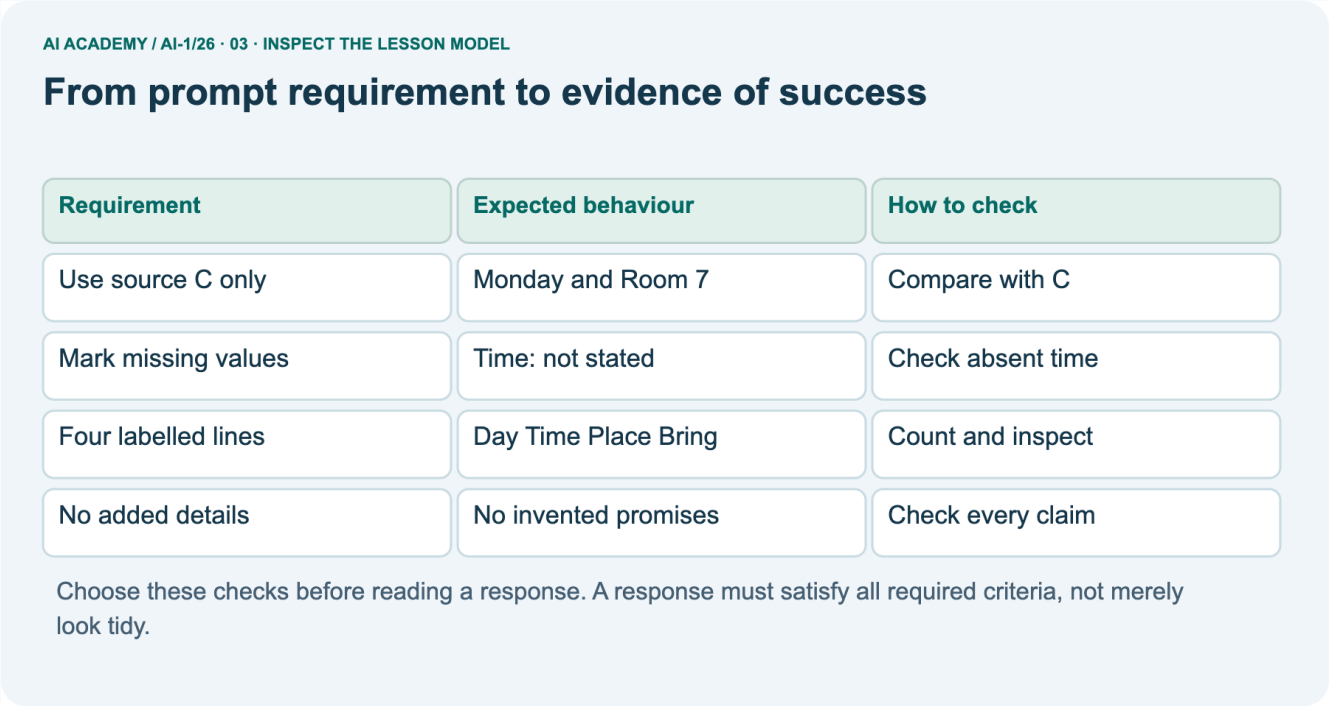
Follow the labelled picture with the matching explanation. Open the vector diagram

State task: Produce exactly two labelled lines using source A only. → Find **evidence**: A supplies Paper Birds but no duration. → Follow missing rule: Exhibit: Paper Birds.

Duration: not supplied. → Check contract: Check labels, two-line format and each factual claim.

> Point to each numbered stage and explain which recorded **input** it uses.

03. From prompt requirement to evidence of success



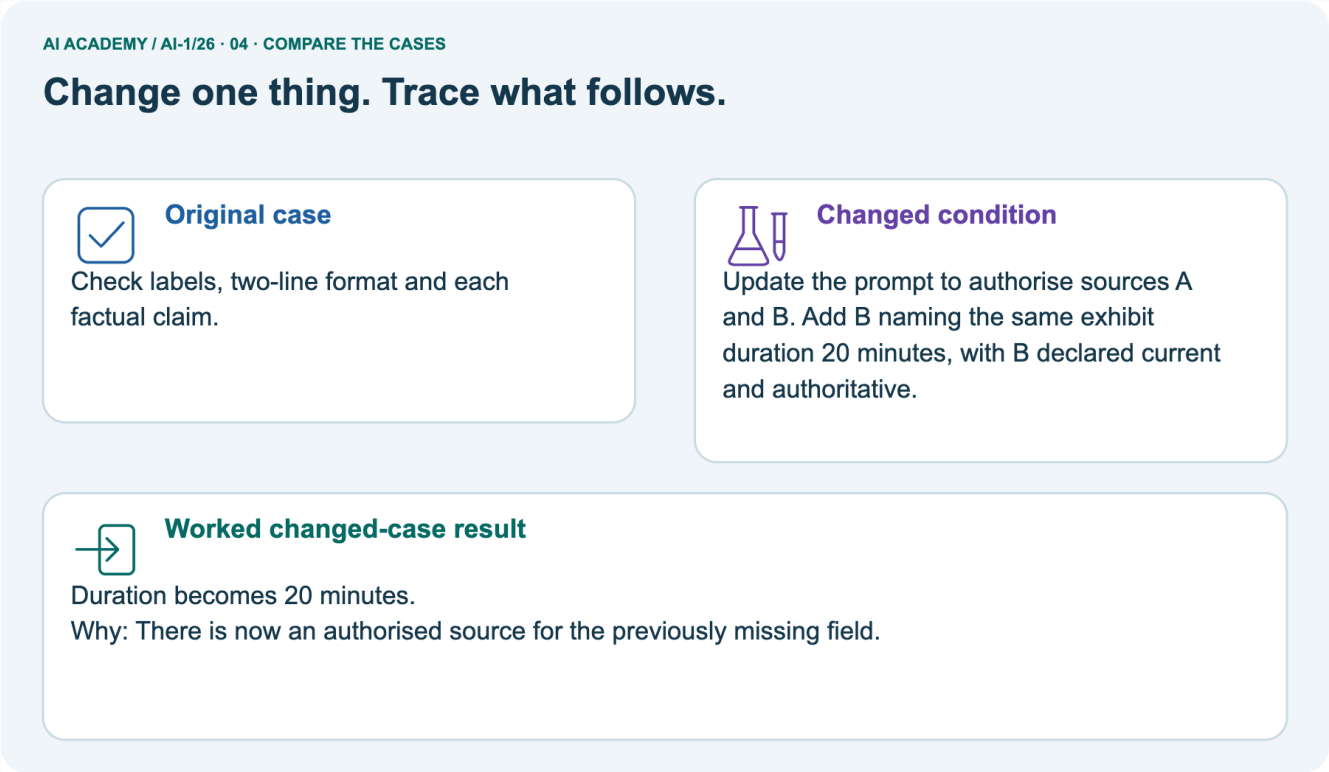
Follow the labelled picture with the matching explanation. Open the vector diagram

From **prompt** requirement to **evidence** of success. Use source C only | Monday and Room 7 | Compare with C. Mark missing values | Time: not stated | Check absent time.

Four labelled lines | Day Time Place Bring | Count and inspect. No added details | No invented promises | Check every claim

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



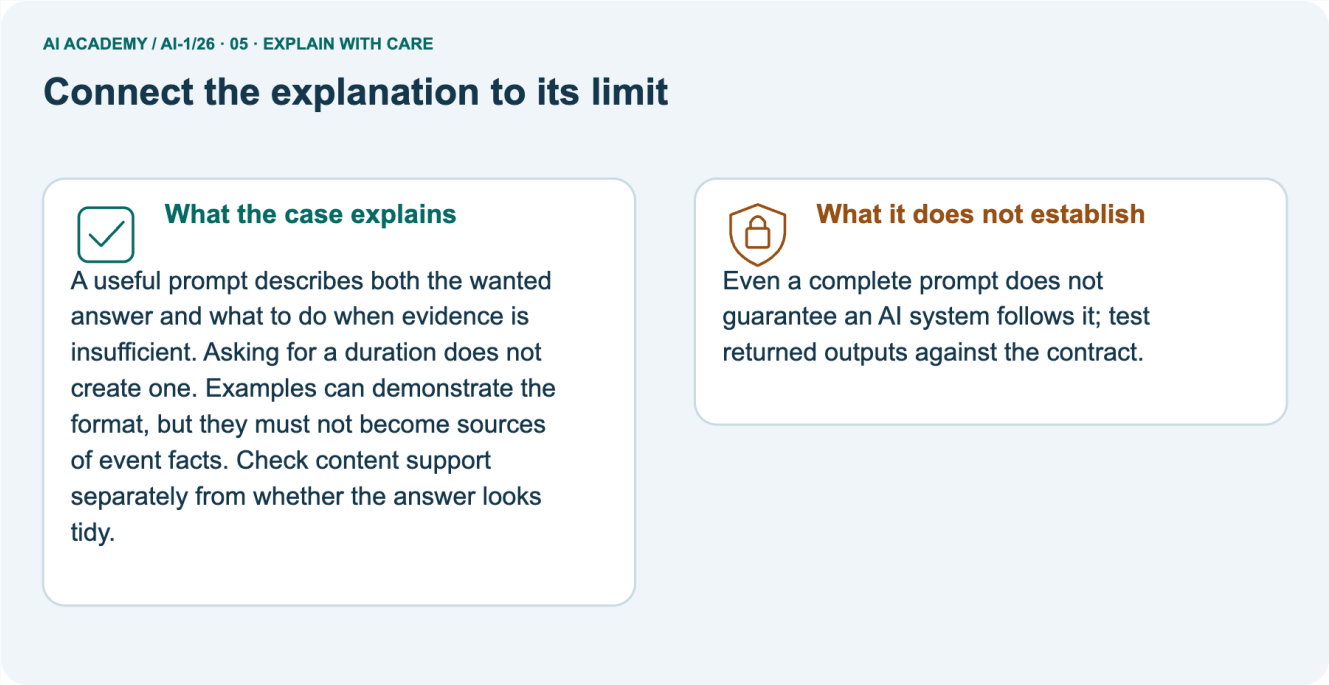
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Check labels, two-line format and each factual claim. Changed condition: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Expected result: Duration becomes 20 minutes. Reason: There is now an authorised source for the previously missing field.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



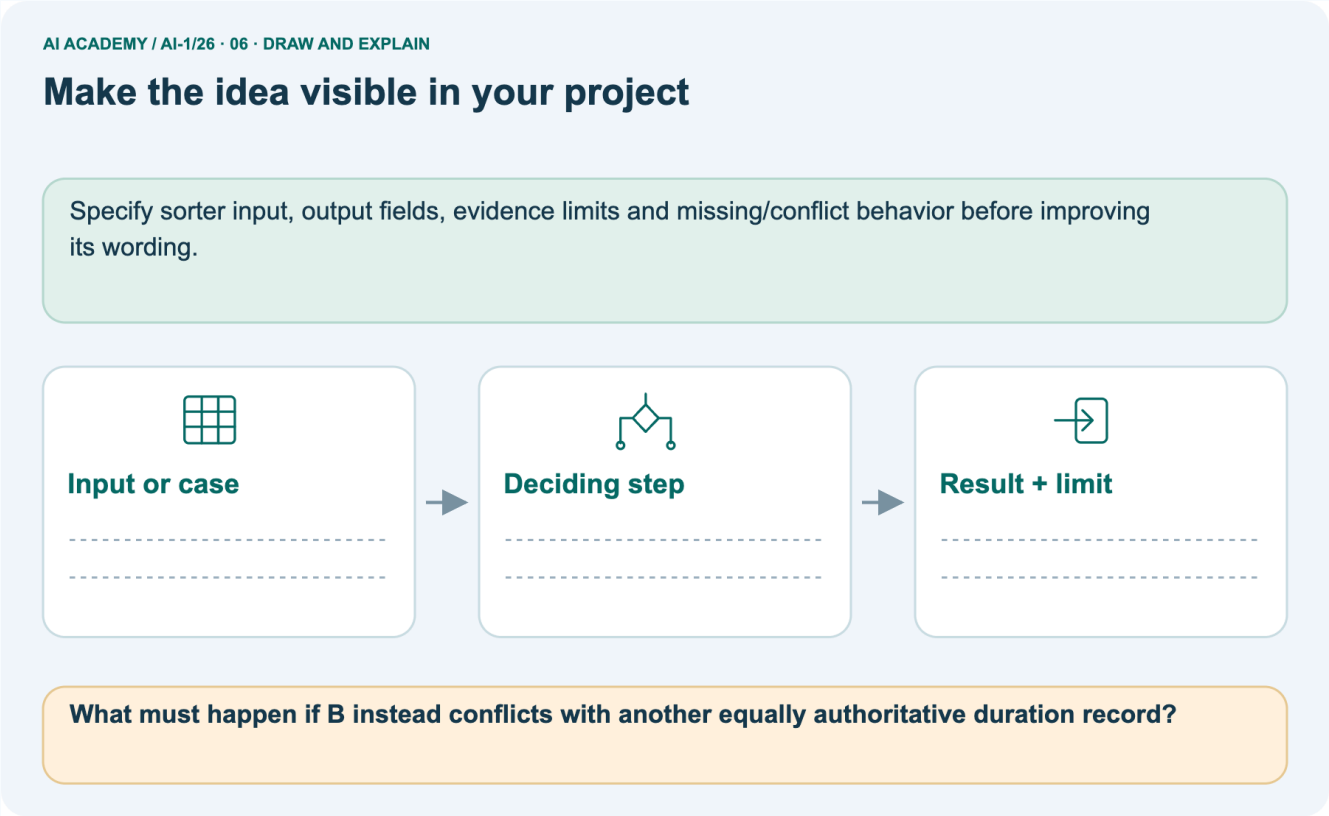
Follow the labelled picture with the matching explanation. Open the vector diagram

A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy. Limit: Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording. Draw your **input**, deciding step, result and limit.

Supported question: What must happen if B instead conflicts with another equally authoritative duration record?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.